



Ecosystem Reputation: A Non-Transferable Trust Primitive for Conservation Contributors

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Abstract

Token-weighted governance is plutocratic: those who buy the most tokens dominate decisions regardless of competence or commitment. We present Zoo Network’s Ecosystem Reputation System, implementing ZIP-0026, a soulbound token (SBT) based trust primitive that introduces a meritocratic dimension to conservation governance. Non-transferable ZOO-REP tokens are issued to contributors based on verified activities: governance participation, grant completion, peer review, code contributions, citizen science observations, and conservation field work. Reputation accumulates across six categories, decays at 5% annually during inactivity, and translates to governance weight multipliers (1.0x–2.5x) that amplify the voting power of active contributors relative to passive token holders. We formalize the reputation scoring function, prove convergence properties of the decay mechanism, analyze anti-gaming defenses, and describe the **SBTReputation** smart contract. Deployment on Zoo Network demonstrates that reputation weighting increases governance participation by 34% and shifts funded proposals toward higher-quality conservation outcomes as measured by post-funding impact scores.

Keywords: soulbound tokens, reputation, non-transferable, governance, conservation, anti-Sybil

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1 Introduction

The original vision of decentralized governance assumed that token holders would be informed, engaged participants whose economic stake aligned with protocol health. In practice, governance is dominated by a small number of large holders (“whales”) whose interests may diverge from the broader community [Barbereau et al.(2022)]. In DeFi governance, the top 10 addresses frequently control over 50% of voting power [Barbereau et al.(2022)]. This plutocratic structure is especially problematic for conservation governance, where the people with the most relevant knowledge—field researchers, citizen scientists, local conservationists—often have the smallest token holdings.

Reputation systems address this by credentialing contributors based on demonstrated activity rather than capital. Soulbound tokens (SBTs), proposed by Weyl, Ohlhaver, and Buterin [Weyl et al.(2022)], provide the on-chain primitive: non-transferable tokens that represent credentials, affiliations, and reputation. Unlike transferable tokens, SBTs cannot be bought, sold, or speculated on, making them resistant to market capture.

Zoo Network’s Ecosystem Reputation System implements ZIP-0026 as a comprehensive on-chain reputation framework for conservation governance. ZOO-REP tokens are soulbound, category-aware, decay-subject, and governance-integrated.

1.1 Contributions

1. A **multi-category reputation model** with six contribution types and category-specific scoring (Section 3).
2. **SBTReputation smart contract** implementing ERC-5192 (Minimal Soulbound NFTs) with governance weight computation (Section 4).
3. A **formal decay model** with convergence analysis (Section 5).
4. **Anti-gaming mechanisms** including issuer rate limits, anomaly detection, and reputation revocation (Section 6).
5. **Governance integration** showing how reputation weights interact with token-weighted voting (Section 7).
6. **Empirical evaluation** of reputation effects on governance quality (Section 8).

2 Related Work

2.1 Soulbound Tokens

Weyl, Ohlhaver, and Buterin [Weyl et al.(2022)] proposed SBTs as non-transferable tokens encoding social identity, credentials, and affiliations. EIP-5192 [EIP-5192(2022)] standardizes the minimal SBT interface. EIP-4973 [EIP-4973(2022)] defines account-bound tokens with consent-based issuance. Zoo uses ERC-5192 for compatibility with standard wallets and block explorers.

2.2 On-Chain Reputation

SourceCred [SourceCred(2020)] measures contributions to open-source projects through a graph-based algorithm. Coordinape [Coordinape(2021)] uses peer attestation for DAO compensation. Gitcoin Passport [Gitcoin Passport(2023)] aggregates identity signals for Sybil resistance. These

systems focus on software communities; Zoo extends the approach to conservation with field work verification and ecological impact measurement.

2.3 Reputation in Governance

Optimism’s Citizen House uses badge-based governance for retroactive funding decisions [Optimism(2023)]. Lido’s dual governance separates stETH holder voting from node operator input [Lido(2023)]. Zoo’s approach is distinct: reputation multiplies rather than replaces token voting power, creating a continuous spectrum between pure plutocracy and pure meritocracy.

2.4 Reputation Decay

PageRank [Page et al.(1999)] applies damping to prevent unbounded reputation accumulation. Stackelberg attention allocation models [Jackson(2008)] suggest that reputation should reflect recent rather than lifetime contributions. Zoo implements exponential decay with activity reset, following similar patterns in academic h-index variants [Hirsch(2005)].

3 Reputation Model

3.1 Contribution Categories

ZOO-REP accumulates across six categories, each representing a distinct form of ecosystem contribution:

Table 1: Reputation categories and earning rates.

Category	Activity	Points	Verification
Governance	Vote cast	1	On-chain
Governance	Proposal created (passed)	10	On-chain
Review	Grant reviewed	5	Council attestation
Review	Peer review completed	10	Review contract
Grants	Grant completed (700+)	50	Impact Oracle
Grants	Grant completed (900+)	100	Impact Oracle
Code	PR merged	10	Technical Council
Code	Critical bug bounty	100	Security Council
Field Work	Data submitted	2	Oracle validation
Field Work	Species observation	1	iNaturalist API
Community	Education event	15	Community Council
Community	Oracle operation (monthly)	5	Automated

3.2 Tier System

Total reputation maps to governance tiers:

The weight multiplier is applied to token-based voting power:

$$\text{EffectiveVotes}(a) = \text{TokenBalance}(a) \times \text{TierMultiplier}(\text{Rep}(a)) \quad (1)$$

An Elder with 1,000 ZOO has the same governance power as an Observer with 2,500 ZOO.

Table 2: Reputation tiers and governance benefits.

Tier	Points	Weight	Benefits
Observer	0–49	1.0x	Basic participation
Contributor	50–199	1.2x	Grant eligibility
Steward	200–499	1.5x	Council nomination
Guardian	500–999	2.0x	Expedited grant review
Elder	1,000+	2.5x	Council appointment

3.3 Formal Scoring

The total reputation score for address a at time t is:

$$\text{Rep}(a, t) = \sum_{c \in \mathcal{C}} \sum_{k=1}^{n_{a,c}} p_{a,c,k} \cdot D(t - t_{a,c,k}) \quad (2)$$

where $\mathcal{C}$ is the set of categories, $n_{a,c}$ is the number of reputation-earning events for a in category c , $p_{a,c,k}$ is the points for event k , and $D(\Delta t)$ is the decay function applied to events older than the grace period.

4 SBTReputation Contract

```

1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  import "@openzeppelin/contracts/token/ERC721/ERC721.sol";
5
6  contract SBTReputation is ERC721 {
7      struct ReputationData {
8          uint256 totalPoints;
9          uint256 lastActivityTimestamp;
10         mapping(bytes32 => uint256) categoryPoints;
11     }
12
13     enum Tier {
14         Observer, Contributor, Steward,
15         Guardian, Elder
16     }
17
18     mapping(address => ReputationData) internal _rep;
19     mapping(address => bool) public authorizedIssuers;
20     mapping(address => uint256)
21         public monthlyIssued;
22     mapping(address => uint256)
23         public monthlyIssueTimestamp;
24
25     uint256 public constant MAX_MONTHLY_PER_ISSUER = 500;
26     uint256 public constant DECAY_RATE_BPS = 500;
27     uint256 public constant GRACE_PERIOD = 180 days;
28     uint256 public constant DECAY_INTERVAL = 365 days;
29
30     uint256[5] public tierThresholds =

```

```

31         [0, 50, 200, 500, 1000];
32     uint256[5] public tierWeights =
33         [100, 120, 150, 200, 250];
34
35     event ReputationAwarded(
36         address indexed account,
37         bytes32 category,
38         uint256 points,
39         bytes32 evidenceHash
40     );
41     event ReputationSlashed(
42         address indexed account,
43         uint256 amount,
44         string reason
45     );
46     event TierChanged(
47         address indexed account,
48         Tier oldTier,
49         Tier newTier
50     );
51
52     modifier onlyAuthorizedIssuer() {
53         require(
54             authorizedIssuers[msg.sender],
55             "not_authorized_issuer"
56         );
57         _;
58     }
59
60     function awardReputation(
61         address account,
62         bytes32 category,
63         uint256 points,
64         bytes32 evidenceHash
65     ) external onlyAuthorizedIssuer {
66         _enforceMonthlyLimit(msg.sender, points);
67         _applyDecay(account);
68
69         Tier oldTier = getTier(account);
70         _rep[account].totalPoints += points;
71         _rep[account].categoryPoints[category] += points;
72         _rep[account].lastActivityTimestamp =
73             block.timestamp;
74         Tier newTier = getTier(account);
75
76         emit ReputationAwarded(
77             account, category, points, evidenceHash
78         );
79         if (oldTier != newTier) {
80             emit TierChanged(
81                 account, oldTier, newTier
82             );
83         }
84     }
85
86     function slashReputation(
87         address account,
88         uint256 amount,
89         string calldata reason

```

```

90     ) external onlyGovernance {
91         _rep[account].totalPoints =
92             _rep[account].totalPoints > amount
93             ? _rep[account].totalPoints - amount
94             : 0;
95         emit ReputationSlashed(
96             account, amount, reason
97         );
98     }
99
100     function getGovernanceWeight(address account)
101         external view returns (uint256)
102     {
103         uint256 points = _getDecayedPoints(account);
104         Tier tier = _tierFromPoints(points);
105         return tierWeights[uint256(tier)];
106     }
107
108     function getTier(address account)
109         public view returns (Tier)
110     {
111         uint256 points = _getDecayedPoints(account);
112         return _tierFromPoints(points);
113     }
114
115     function getCategoryPoints(
116         address account, bytes32 category
117     ) external view returns (uint256) {
118         return _rep[account].categoryPoints[category];
119     }
120
121     function _getDecayedPoints(address account)
122         internal view returns (uint256)
123     {
124         uint256 points = _rep[account].totalPoints;
125         uint256 lastActive =
126             _rep[account].lastActivityTimestamp;
127         if (lastActive == 0) return points;
128
129         uint256 elapsed =
130             block.timestamp - lastActive;
131         if (elapsed <= GRACE_PERIOD) return points;
132
133         uint256 decayPeriods =
134             (elapsed - GRACE_PERIOD) / DECAY_INTERVAL;
135         for (uint256 i = 0; i < decayPeriods; i++) {
136             points = points *
137                 (10000 - DECAY_RATE_BPS) / 10000;
138         }
139         return points;
140     }
141
142     function _tierFromPoints(uint256 points)
143         internal view returns (Tier)
144     {
145         if (points >= tierThresholds[4])
146             return Tier.Elder;
147         if (points >= tierThresholds[3])
148             return Tier.Guardian;

```



```

149         if (points >= tierThresholds[2])
150             return Tier.Steward;
151         if (points >= tierThresholds[1])
152             return Tier.Contributor;
153         return Tier.Observer;
154     }
155
156     function _enforceMonthlyLimit(
157         address issuer, uint256 points
158     ) internal {
159         uint256 currentMonth =
160             block.timestamp / 30 days;
161         if (monthlyIssueTimestamp[issuer] <
162             currentMonth) {
163             monthlyIssued[issuer] = 0;
164             monthlyIssueTimestamp[issuer] =
165                 currentMonth;
166         }
167         require(
168             monthlyIssued[issuer] + points <=
169                 MAX_MONTHLY_PER_ISSUER,
170             "monthly_issuance_limit"
171         );
172         monthlyIssued[issuer] += points;
173     }
174
175     // Soulbound: disable transfers
176     function _update(
177         address to,
178         uint256 tokenId,
179         address auth
180     ) internal override returns (address) {
181         require(
182             auth == address(0) || to == address(0),
183             "soulbound: non-transferable"
184         );
185         return super._update(to, tokenId, auth);
186     }
187
188     function _applyDecay(address account) internal {
189         _rep[account].totalPoints =
190             _getDecayedPoints(account);
191         _rep[account].lastActivityTimestamp =
192             block.timestamp;
193     }
194 }

```

Listing 1: SBTReputation contract

5 Decay Model

5.1 Decay Function

Reputation decays exponentially during periods of inactivity:

$$D(\Delta t) = \begin{cases} 1 & \text{if } \Delta t \leq \tau_g \\ (1 - r)^{\lfloor (\Delta t - \tau_g) / \tau_d \rfloor} & \text{if } \Delta t > \tau_g \end{cases} \quad (3)$$

where $\tau_g = 180$ days (grace period), $\tau_d = 365$ days (decay interval), and $r = 0.05$ (5% annual decay rate).

5.2 Convergence

Theorem 5.1 (Decay Convergence). *Under continuous inactivity, reputation converges to zero:*

$$\lim_{n \rightarrow \infty} P_0 \cdot (1 - r)^n = 0 \quad \text{for } 0 < r < 1 \quad (4)$$

Proof. This follows directly from the geometric series convergence: $|1 - r| < 1$ for $0 < r < 1$, so $(1 - r)^n \rightarrow 0$ as $n \rightarrow \infty$. $\square$

5.3 Half-Life

The reputation half-life (time to lose 50% of inactive reputation after the grace period) is:

$$t_{1/2} = \tau_g + \tau_d \cdot \frac{\ln 2}{\ln(1/(1 - r))} = 180 + 365 \cdot \frac{0.693}{0.0513} \approx 180 + 4,929 \approx 5,109 \text{ days} \quad (5)$$

At 5% annual decay, reputation halves approximately every 13.5 years of complete inactivity (after the 6-month grace period). This is intentionally slow: conservation researchers may have seasonal activity patterns, and the system should not penalize legitimate intermittent contribution.

5.4 Activity Reset

Any qualifying activity resets the decay timer. The grace period restarts from the most recent activity timestamp:

$$\text{lastActivity}(a) \leftarrow t_{\text{now}} \quad \text{upon any earning event} \quad (6)$$

This means that a contributor who earns even 1 point per year maintains full reputation.

6 Anti-Gaming Mechanisms

6.1 Threat Model

6.2 Issuer Rate Limiting

Each authorized issuer can award at most 500 reputation points per month. This prevents a compromised issuer from inflating reputation rapidly. For context, 500 points is enough to advance from Observer to Steward but not to Guardian. An attacker controlling a single issuer for a full year could award 6,000 points—significant but detectable through anomaly monitoring.

6.3 Multi-Issuer Requirement

Awards exceeding 50 points per event (e.g., critical bug bounties at 100 points) require attestation from two authorized issuers. This prevents unilateral large awards.

Table 3: Reputation gaming threats and defenses.

Threat	Description	Defense
Sybil farming	Create many accounts, earn minimum reputation each	ZooRep gating + on-chain identity correlation
Issuer capture	Compromise an authorized issuer to award arbitrary points	Monthly limit of 500 points per issuer
Vote farming	Cast meaningless votes to accumulate governance REP	Minimum proposal stake prevents trivial proposals
Review gaming	Accept reviews, submit low-quality work for points	Review quality scoring with slashing
Observation spam	Submit false citizen science data	Oracle validation + iNaturalist API cross-check
Collusion	Issuers and recipients coordinate	Multi-issuer requirement for large awards

6.4 Statistical Anomaly Detection

Off-chain monitoring detects unusual patterns:

- Accounts earning points across all categories simultaneously (natural contributors specialize).
- Accounts earning points at exact intervals (humans have irregular contribution patterns).
- Clusters of accounts earning identical points from the same issuer in the same block.

Flagged accounts are reviewed by the Ethics Committee. Confirmed gaming results in reputation slashing (governance vote required: 10% quorum, 66% approval).

6.5 Revocation

Reputation can be reduced or fully revoked for misconduct:

- Fraudulent conservation data: -100 points + 6-month earning freeze.
- Undisclosed conflicts in grant review: -50 points.
- Malicious governance proposals: -100 points.
- Security violations: Full revocation.

7 Governance Integration

7.1 Voting Power Composition

The ZooGovernor contract (companion paper on ZooGov DAO OS) computes effective voting power as:

$$\text{VP}(a) = \text{Tokens}(a) \times \frac{W(\text{Tier}(a))}{100} \quad (7)$$

where W returns the tier weight in basis points ($100 = 1.0x$, $250 = 2.5x$). The division by 100 normalizes to a multiplier.

7.2 Grant Voting Weight

In quadratic funding rounds, contributor ZooRep modifies the matching formula:

$$M_j^{\text{rep}} = \left(\sum_{i=1}^{n_j} \sqrt{c_{ij}} \cdot \sqrt{W_i/100} \right)^2 - \sum_{i=1}^{n_j} c_{ij} \quad (8)$$

This amplifies matching for grants supported by high-reputation contributors relative to grants supported only by token holders with no track record.

7.3 Council Eligibility

Reputation tiers gate council eligibility:

- **Community Council:** Contributor tier (≥ 50 REP) to nominate, Steward (≥ 200) to serve.
- **Research Council:** Steward to nominate, Guardian (≥ 500) to serve.
- **Emergency Council:** Elder ($\geq 1,000$) required.

8 Empirical Evaluation

8.1 Reputation Distribution

After 12 months of operation:

Table 4: Reputation distribution across tiers.

Tier	Accounts	% of Active	Median Points
Observer	2,341	67.2%	12
Contributor	743	21.3%	97
Steward	284	8.2%	312
Guardian	89	2.6%	674
Elder	27	0.8%	1,842

8.2 Governance Impact

Reputation weighting increased participation by 34%, reduced median voter token balance (indicating broader participation), lowered the Gini coefficient of voting power from 0.78 to 0.54, and improved mean funded proposal impact by 18%.

Table 5: Governance metrics before and after reputation weighting.

Metric	Before REP	After REP
Unique voters per proposal	47	63 (+34%)
Median voter token balance	12,400	3,200
Gini of voting power	0.78	0.54
Funded proposal impact score	612	724 (+18%)

9 Security Considerations

Soulbound enforcement. The `_update` override prevents token transfers. Reputation can only be earned through authorized issuers and lost through governance-approved slashing. There is no secondary market for ZOO-REP.

Issuer key management. Authorized issuers (council contracts, oracle contracts) use multi-sig or governance-controlled addresses. A compromised issuer can award at most 500 points per month before detection.

Tier threshold manipulation. Tier thresholds are governance parameters (Meta-Governance proposal type: 20% quorum, 75% approval). Lowering thresholds to inflate Elder count requires broad community consensus.

Privacy. Reputation scores are public. Contributors who need privacy use pseudonymous addresses. The system does not link reputation to real-world identity; decoupling is by design.

Centralization risk. Initially, Foundation-appointed issuers dominate. The path to full decentralization: progressive transfer of issuer authority to DAO-elected councils as governance matures.

10 Conclusion

The Zoo Ecosystem Reputation System introduces a meritocratic trust primitive that complements token-weighted governance. By issuing non-transferable reputation tokens across six contribution categories, the system rewards sustained, verified activity in conservation, governance, development, and community building. The 5% annual decay ensures that reputation reflects current engagement. Governance weight multipliers (up to 2.5x) give active contributors meaningful influence without entirely overriding economic stake. Empirical results demonstrate 34% higher governance participation, reduced voting power concentration (Gini $0.78 \rightarrow 0.54$), and 18% improvement in funded proposal quality when reputation weighting is active. The system provides a concrete implementation of the soulbound token thesis for scientific governance.

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